INCH–POUND

MIL–DTL–51194D(EA)
8 July 2003

SUPERSEDING
MIL–F–51194C(EA)
6 September 1990

DETAIL SPECIFICATION

FILTER, PARTICULATE, 20 CFM, M19

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of particulate filter rated at 20 standard cubic feet per minute airflow.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of De-

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technical Director, U.S. Army Edgewood Chemical Biological Center, ATTN: AMSSB–REN–SE–SS, Aberdeen Proving Ground, MD 21010–5424 by using the self–addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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fense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

DEPARTMENT OF DEFENSE

MIL–STD–282 – Filter Units, Protective Clothing, Gas–Mask Components and Related Products: Performance – Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111–5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY EDGEWOOD CHEMICAL BIOLOGICAL CENTER

DRAWINGS

5–19–1175 – Filter, Particulate, 20 CFM, M19 Assembly

(Copies are available from Technical Director, U.S. Army Edgewood Chemical Biologcal Center, ATTN: AMSSB–REN–SE, Aberdeen Proving Ground, MD 21010–5424.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials, components, and assembly.

3.1.1 Materials. All materials cited on Drawing 5–19–1175 and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the particulate filter shall conform to the specifications and drawings listed on Drawing 5–19–1175 and subsidiary drawings.

3.1.3 Assembly. The particulate filter shall be assembled as specified on Drawing 5–19–1175.

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3.2 Airflow resistance. The resistance to airflow shall be no greater than 1.25 inches of water gage (iwg) at an airflow rate of 20 cubic feet per minute (cfm) at standard conditions (70° F and 1013 mbar).

3.3 Filtration efficiency. The filter shall be no less than 99.97% efficient in the removal of aerosols with a mean diameter of 0.3 microns.

3.4 Resistance to rough handling. The filter shall meet airflow resistance, filtration efficiency, and air leakage requirements after rough handling for 15 minutes at an amplitude of 3/4 inch with the filter face, pleats, and separators in a vertical position.

3.5 Air leakage. Air leakage (in and around gasket area) shall be not greater than 0.91 cubic inches per minute when the filter is internally pressurized to a pressure of not less than 12 inches of water.

3.6 Workmanship. The filter shall be free from chipped, burred, or bent metal sections and loose or damaged gaskets; foreign matter, such as oil or viscous substances, and rips, tears, or holes in the filtering medium.

3.7 First article. When specified (see 6.2), a sample of filters shall be subjected to first article inspection in accordance with 4.2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article sample shall consist of 20 particulate filters manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.2 Inspections to be performed. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings.

4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The

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contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

TABLE I. First article requirement verification matrix

Characteristic	Requirement Paragraph	Verification Paragraph
Materials, components, and assembly	3.1	4.4.1
Airflow resistance	3.2	4.4.2
Filtration efficiency	3.3	4.4.3
Rough handling	3.4	4.4.4
Air leakage	3.5	4.4.5
Workmanship	3.6	4.4.6

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the filters produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for acceptance at one time; however, no more than one lot of filter material shall be represented in any one lot of finished filters. Each lot shall be identified by an alphanumeric lot number. The lot number shall include a manufacturer's identification symbol consisting of 3 alpha characters, a numeric code identifying the year of production, a code or abbreviation that signifies the month of production, and an interfix–serial number. The interfix–serial number shall change if there is a change in the design, manufacturing process, materials, suppliers, production run, or if a new contract is used.

4.3.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.3.5 and, when specified, table I. Samples shall be selected at random.

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TABLE I. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.3.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. Sample filters shall be examined and tested in accordance with the classification of characteristics in 4.3.5. If a filter is found that does not conform to any characteristic inspected 100%, such as filtration efficiency and airflow resistance, the non–conforming filter shall be rejected and removed from the lot. For characteristics other than filtration efficiency and airflow resistance, failure of any sample filter to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented (see footnote, Table I).

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals who use or maintain the product. Characteristics whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item are also critical characteristics. Major characteristics are characteristics whose nonconformance to

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specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. Unless otherwise specified, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table I for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.3.5	Filter, particulate, 20 CFM, M19		5–19–1175
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Filtration efficiency	100 percent	3.3	4.4.3
2	Air leakage	100 percent	3.5	4.4.5
Major				
101	Airflow resistance	100 percent	3.2	4.4.2
102	Component evident	Table I, Level VII	3.1	4.4.1
103	Component correctly assembled	Table I, Level VII	3.1	4.4.1
104	Marking correct	Table I, Level VII	3.1	4.4.1
105	Overall dimensions correct	Table I, Level VII	3.1	4.4.1
106	Workmanship	Table I, Level VII	3.6	4.4.6
Minor				
201	Marking legible	Table I, Level IX	3.1	4.4.1

NOTES:

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4.4 Verification methods and procedures.

4.4.1 Material, components, and assembly. Verify by examination of filter and drawings. Examination involves inspection of the item using the un-aided eye, tools, gages, or other commercial measuring devices.

4.4.2 Airflow resistance. The airflow resistance shall be determined at the airflow of 20 cfm. The measured pressure drop across the particulate filter, when corrected to standard conditions of 21° C (70° F) and 1 atm (1013 mbar), shall be no greater than that specified in 3.2. The up–stream and down–stream static pressure measuring tubes shall be as close as possible to the filter and shall not be on a section of duct that has a changing cross sectional area. Test for airflow resistance as follows:

(a) Connect the filter to a source of forced air.
(b) Set the flow of air through the filter to 20 cfm.
(c) Record the barometric pressure.
(d) Measure and record the air stream temperature.
(e) Determine and record the difference between up–stream and downstream static pressure.

If the recorded air stream temperature and barometric pressure is different than 21° C (70° F) and 1 atm, calculate and record the air flow resistance (ΔP (cal)) using the equation below to correct the measured airflow resistance to airflow resistance at standard conditions:

$$\Delta P\ (cal) = \Delta P\ (measured) \times P\ (test) \times 86.21 \div [(492 + 1.8T)^{1.768}]$$

where:

Δ P (cal) – air flow resistance corrected to standard conditions in millimeters of water gage (mm wg).
P (test) – barometric pressure at time of test in millimeters of mercury (mm Hg).
Δ P (measured) – air flow resistance from test measurement in millimeters of water.
T – temperature of air stream flowing through the filter in degrees C.

Note: Correction for nonstandard conditions is not required when it is clear that the passage or failure of the filter is not in doubt.

4.4.3 Filtration efficiency. The test apparatus used to measure the filtration efficiency shall be capable of maintaining a stable concentration of aerosol with a mean particle size of 0.3 microns with a steady flow through the filter of 20 scfm (see 6.4). The test apparatus shall have the capability to determine the concentration of aerosol both upstream and downstream of the filter within 0.005%. The downstream sample point shall be located such that changing of its location across the ductwork does not cause a significant change in measured concentrations. This shall be verified using a filter with a known measurable leak. The filter shall be placed in the test apparatus such that the flow of air containing the challenge aerosol is in the direction indicated by the marking on the filter, and the flow of aerosolized air shall be maintained at 20 ± 3

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scfm. This flow shall be maintained for not less than 1 minute or more than 2 minutes. Filtration efficiency is calculated as from the concentrations or particle counts as follows:

Penetration % = (concentration downstream of the filter/concentration upstream) x 100, or
Penetration % = (no. particles counted downstream/ no. of particles counted upstream) x 100

Filtration efficiency % = 1 – penetration %

4.4.4 Resistance to rough handling. Fifteen (15) first article filters, which have previously passed airflow resistance (4.4.2) and filtration efficiency (4.4.3) tests, shall be subjected to rough handling in accordance with method T105.10 of MIL–STD–282. Then 5 each of the 15 filters shall be tested for resistance to airflow (4.4.2), filtration efficiency (4.4.3) and air leakage (4.4.5).

4.4.5 Air leakage. The sample filters shall be tested for leakage using a test apparatus which shall have two flat (within 0.005 inch) smooth (32 micro inch) surfaced metal plates at least 0.25 inch thick that can be clamped to the gaskets on the inlet and outlet of the filter. The clamping mechanism shall be controlled so that the total combined compression of the two gaskets does not exceed 0.120 inch. There shall be provisions in one of the plates which shall be fixed to the base of the apparatus for pressurized air to be introduced into the filter under test through a flow meter capable of measuring a flow of 0.91 ± 0.06 cubic inches per minute of air flow. There shall also be provisions for measuring the pressure inside the filter between these plates through the use of a manometer or pressure gage capable of measuring pressures of 12.00 inches of water with an accuracy of ±0.06 inches of water. The internal pressure measurement shall be taken through a separate passage through one of the two plates from the passage that is used to introduce the measured flow of air. The apparatus shall also have a flow regulation valve by which the air flow can be controlled. The test shall be conducted by clamping the filter to be tested between the two metal plates, opening the air flow control valve, and adjusting the flow until a pressure of at least 12 inches of water is established within the filter. The flow rate that is required to maintain this pressure is the air leakage value.

4.4.6 Workmanship. Verify by examination.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the man-

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aging Military Department's or Defense Agency's automated packaging files, CD–ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The particulate filter covered by this specification is intended for use in the M13A1 gas–particulate filter units.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)
(c) First article:
  (1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
  (2) Name and address of test facility and shipping instructions when testing is performed by the Government.
  (3) Time required for the Government to notify the contractor whether or not to proceed with production.
(d) Shelf life codes

6.3 Submission of alternative inspection provisions. Proposed alternative inspection provisions may be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Filtration efficiency tests and aerosol materials. The TDA 100 Filter Penetration Tester manufactured by Air Techniques Inc. has been found to be capable of performing the filtration efficiency test when operated with Poly–Alpha Olefin (common trade names are Ethyl Flow 164 and Emery 3004) or Dioctylphthalate (DOP) as the aerosol agent. DOP is not recommended because of its potential as a carcinogen.

6.5 Subject term (key word) listing.
Air purifier

Custodian:

Army – EA

Preparing activity

Army – EA
Project No. 4240–A268

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYYYMMDD)*
	MIL-DTL-51194D(EA)	20030708

3. DOCUMENT TITLE

FILTER, PARTICULATE, 20 CFM, M19

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include ZIP Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	**7. DATE SUBMITTED** *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
U.S. Army Edgewood Chemical Biological Center	(1) Commercial (410) 436-3230	(2) DSN 584-3230
c. ADDRESS *(Include ZIP Code)* ATTN: AMSSB-REN-SE-SS, E3331 5183 Blackhawk Road Aberdeen Proving Ground, MD 21010-5424	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888	

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